

Akshat Sureshbhai Desai

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EDUCATION

California State University, Fullerton

Aug 2024 - May 2026

Master of Science in Computer Science

GPA: 3.6/4.0

Relevant Coursework: Neural Networks and Deep Learning, Natural Language Processing, Operating System Design

Charotar University of Science and Technology

Nov 2020 - May 2024

Bachelor of Technology in Computer Science

GPA: 3.6/4.0

Relevant Coursework: Machine Learning, Artificial Intelligence, Data Science & Analytics, Image Processing and CV

PROFESSIONAL EXPERIENCE

Graduate Research Assistant - California State University, Fullerton, CA

Nov 2024 - Present

Project: Serverless AI Tutoring Engine (Verilog HDL)

- **Built a full-stack AI tutoring assistant** for Verilog HDL, featuring a Q&A chatbot, an automated code generator, and an integrated code execution environment for university students.
- **Optimized an end-to-end NLP pipeline** by fine-tuning a custom LLM on a self-generated dataset to accurately process and respond to natural language queries.
- **Implemented a scalable, serverless architecture** on Google Cloud Run, leveraging Docker, StreamLit, LangChain, and ChromaDB for optimized performance.

Project: HPC Medical Imaging Pipeline (Alzheimer's Disease)

- **Constructed an end-to-end ML classification pipeline** for the early detection of Alzheimer's disease and Mild Cognitive Impairment (MCI) using brain MRI Radiomics and Hippocampus data.
- Extracted the ADNI dataset by leveraging FreeSurfer, PyRadiomics, NiBabel, and itk-SNAP on HPC clusters (SLURM) to extract complex neuroimaging features.
- **Achieved 92.7% diagnostic accuracy** by implementing an XGBoost Classifier and utilizing scikit-learn (Lasso) for optimized feature selection.

Machine Learning Engineer - Indian Space Research Organisation, Ahmedabad, India

Dec 2023 - May 2024

- **Devised a real-time Haar-Cascade object tracking system** for satellite cameras, utilizing a skewness-based auto-exposure algorithm to observe orbital debris under volatile lighting.
- **Validated the dynamic computer vision pipeline** within a simulated space environment, integrating Python, OpenCV, and NumPy to process live hardware feeds from IDS cameras via the IDS uEye SDK.

PROJECTS

Lead AI Developer - Lightwall (Industry Collaboration), Fullerton, CA

Oct 2025 - Jan 2026

- **Fabricated a real-time AI art installation** for Worthwhile LLC, seamlessly integrating hardware and software to engage observers through two-way voice conversations and motion-responsive lighting.
- **Integrated physical sensors with edge AI** by deploying a local Llama LLM on a Mac mini, integrating radar detection to translate human movement and speech into dynamic artistic expressions.
- **Architected a self-aware RAG pipeline** featuring radar-based personality switching, giving the installation real-time context of its own memory and codebase to dynamically process and reply to audience interactions.

HACKATHONS & AWARDS

CSU AI Summer Camp by AWS, Cal Poly San Luis Obispo, CA

Aug 2025

- **Programmed a predictive dining dashboard ($R^2=0.96$)** for Cal Poly Pomona, utilizing an interactive UI with dynamic visualizations to forecast granular staffing across six categories, projecting \$52K annual savings.
- **Formulated an AI conversational agent** (AWS Bedrock, LangChain) enabling administrators to seamlessly query operational insights, historical trends, and complex prediction logic via natural language.

Skills

Programming Languages: Python, C/C++, SQL, Bash/Shell

AI and Machine Learning: PyTorch, TensorFlow, XGBoost, Scikit-Learn, ONNX, OpenCV, Pandas, NumPy

GenAI and LLM Ecosystem: vLLM, LangChain, LangGraph, Hugging Face, ChromaDB, RAG, LLaMA, YOLO

Cloud and MLOps: AWS (Bedrock), Google Cloud Run, Microsoft Azure, Docker, Linux HPC (SLURM), CUDA/cuDNN

Web and Developer Tools: Flask, Streamlit, Dash, Firebase, Git